Electric Current

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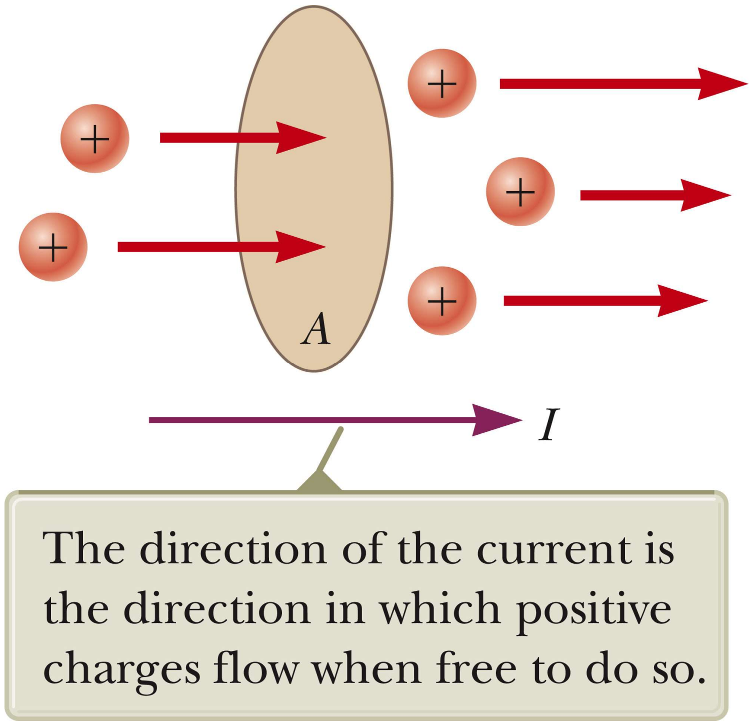
Electric Current

- **Electric current** is the rate of flow of charge through some region of space.
- The SI unit of current is the **ampere** (A).
  - 1 A = 1 C / s
- The symbol for electric current is I.

Electric current

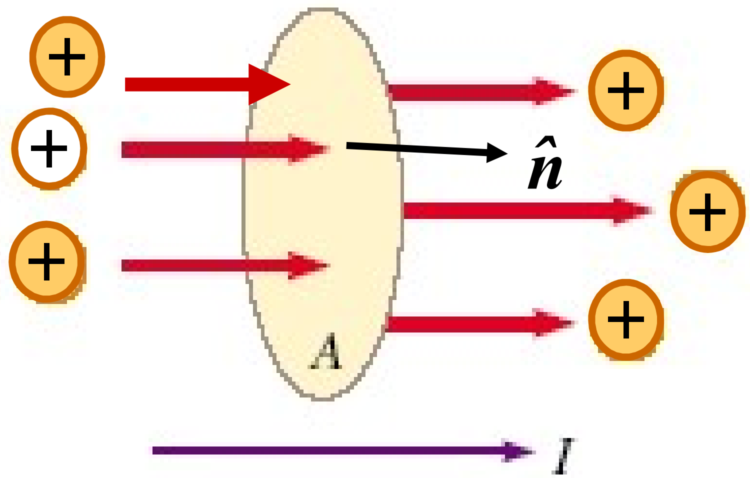
- Electric **current** is defined through a certain **surface**. The current is the rate at which charge flows through the surface. If ΔQ is the amount of charge that passes through an area A (out of a certain side of the area) in a time interval Δt, the average current I_{av} is equal to the charge that passes through A per unit time:

$$I_{av} = \frac{\Delta Q}{\Delta t}$$



Electric current

Current is an algebraic scalar quantity. To determine the sign of current we have to define a certain side of the surface as the side through which current is measured. If positive charges flow out of this side, current will be positive.



Electric Current

- If the rate at which charge flows varies in time, then the current varies in time; we define the instantaneous current *I* as the differential limit of average current:

$$\boxed{I = \frac{dQ}{dt}}$$

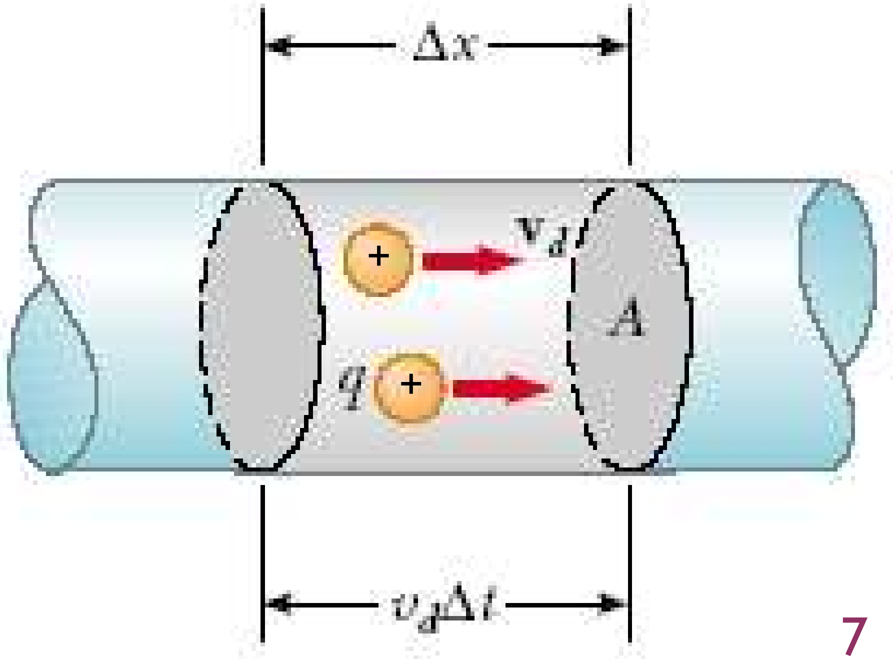
Direction of Current

- The charged particles passing through the surface could be positive, negative or both.
- It is conventional to assign to the current the same direction as the flow of positive charges.
- In an ordinary conductor, the direction of current flow is opposite the direction of the flow of electrons.
- It is common to refer to any moving charge as a *charge carrier*.

Current and Drift Speed

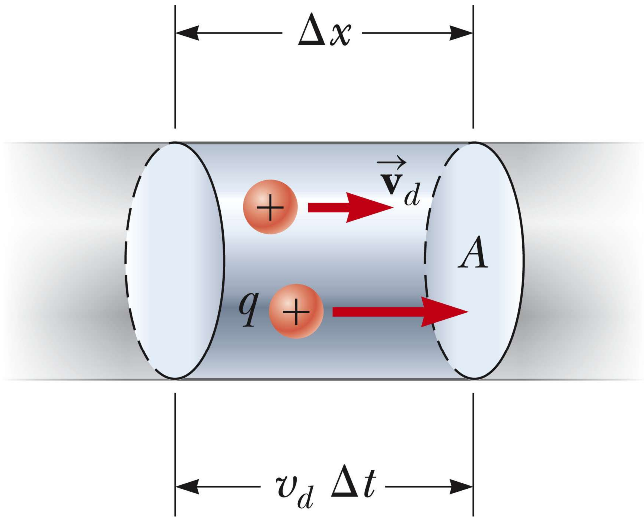
- Charged particles move through a cylindrical conductor of cross-sectional area *A*.
- *n* is the number of mobile charge carriers per unit volume.
- $nA\Delta x$ is the total number of charge carriers in a segment.

$$\Delta Q = (nA\Delta x)q$$



Current and Drift Speed

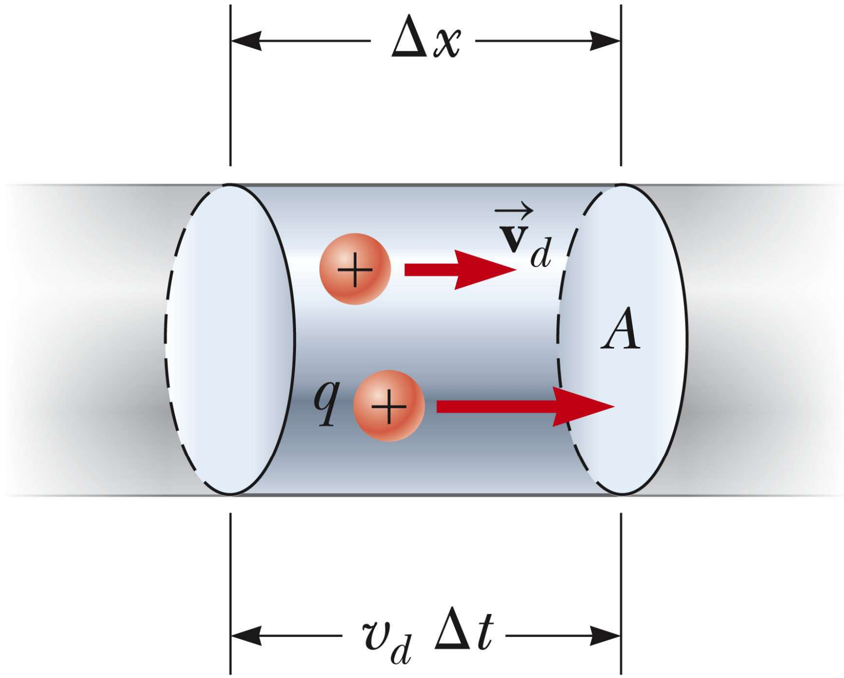
- The total charge is the number of carriers times the charge per carrier, q.
  - $\Delta Q = (nA\Delta x)q$
- Assume the carriers move with a velocity parallel to the axis of the cylinder If v_d is the speed at whicl the carriers move, then
  - $v_d = \Delta x / \Delta t$ and $\Delta x = v_d\, \Delta t$
- v_d is an average speed called the **drift speed.**



Current and Drift Speed

$$\Delta Q = (nA\mathrm{v}_d q)\Delta t$$

$$I = \frac{\Delta Q}{\Delta t} = (nA\mathrm{v}_d q)$$



Motion of Charge Carriers

- In the presence of an electric field, in spite of all the collisions, the charge carriers slowly move along the conductor with a drift velocity,

$$v = \mu E$$

- The electric field exerts forces on the conduction electrons in the wire.
- These forces cause the electrons to move in the wire and create a current.
- μ is the mobility of the material

Motion of Charge Carriers

- The electrons are already in the wire.
- They respond to the electric field set up by the battery.
- The battery does not supply the electrons, it only establishes the electric field.

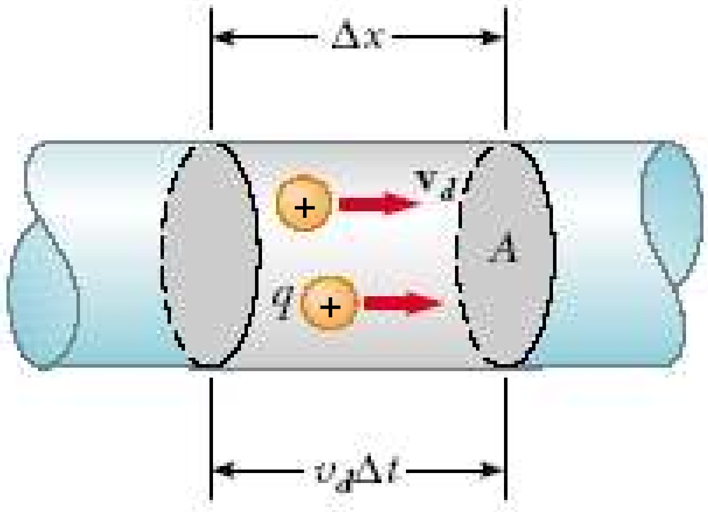
Current Density

- J is the **current density** of a conductor.
- It is defined as the current per unit area.
  - $J \equiv I / A = nq\mathbf{v}_d$
  - This expression is valid only if the current density is uniform and A is perpendicular to the direction of the current.
- J has SI units of A/m^2
- The current density is in the direction of the positive charge carriers.

Current density

- Consider a conductor of cross-sectional area *A* carrying a current *I*. The current density ***J*** in the conductor is defined as the current per unit area. Because the current $I = nqv_dA$ the current density is

$$J \equiv \frac{I}{A} = nqv_d$$



$$I = \int_{\text{the surface}} \boldsymbol{J}.d\boldsymbol{A}$$

Conductivity

- A current density and an electric field are established in a conductor whenever a potential difference is maintained across the conductor.
- For some materials, the current density is directly proportional to the field.
- The constant of proportionality, σ, is called the **conductivity** of the conductor.
- The quantity $\rho = 1/\sigma$ is called the resistivity of the material of the conductor.

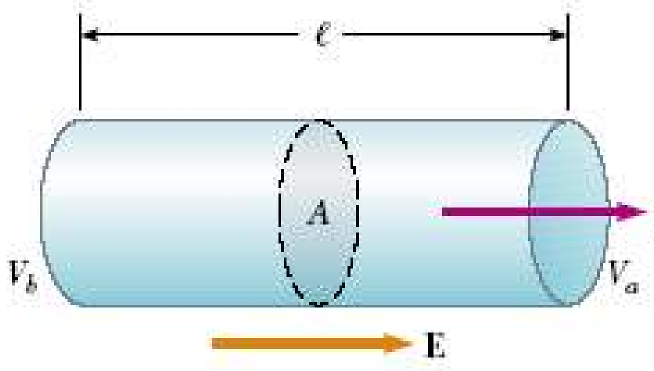
Ohm's Law

- **Ohm's law** states that for many materials, the ratio of the current density to the electric field is a constant σ that is independent of the electric field producing the current.

$$\boxed{J = \sigma E}$$

Ohm's Law

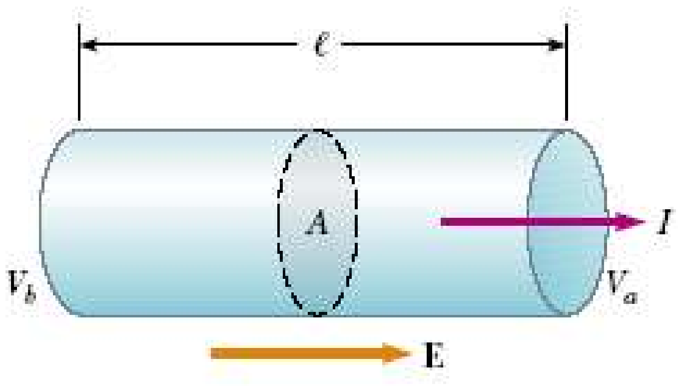
- We can obtain a form of Ohm's law useful in practical applications by considering a segment of straight wire of uniform cross-sectional area *A* and length *l* , as shown in the figure. A potential difference $\Delta V = V_b - V_a$ is maintained across the wire, creating in the wire an electric field and a current. If the field is assumed to be uniform, the potential difference is related to the field through the relationship



$$\Delta V = El \qquad J = \sigma E = \sigma \frac{\Delta V}{l}$$

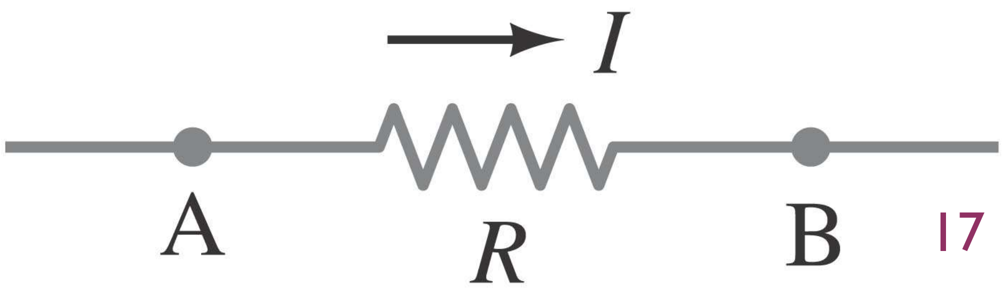
Ohm's Law

$$\therefore \Delta V = \frac{l}{\sigma} J = \frac{l}{\sigma A} I$$



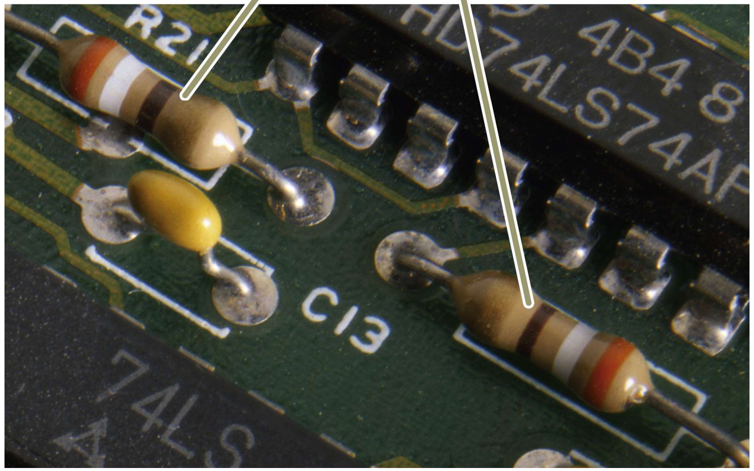
- The quantity ($l/\ \sigma A$) is called the resistance R of the conductor. It is defined as the ratio of the potential difference across a conductor to the current through the conductor:

$$R = \frac{\Delta V}{I} = \frac{l}{\sigma A} = \frac{\rho l}{A}$$



Resistors

The colored bands on these resistors are orange, white, brown, and gold.

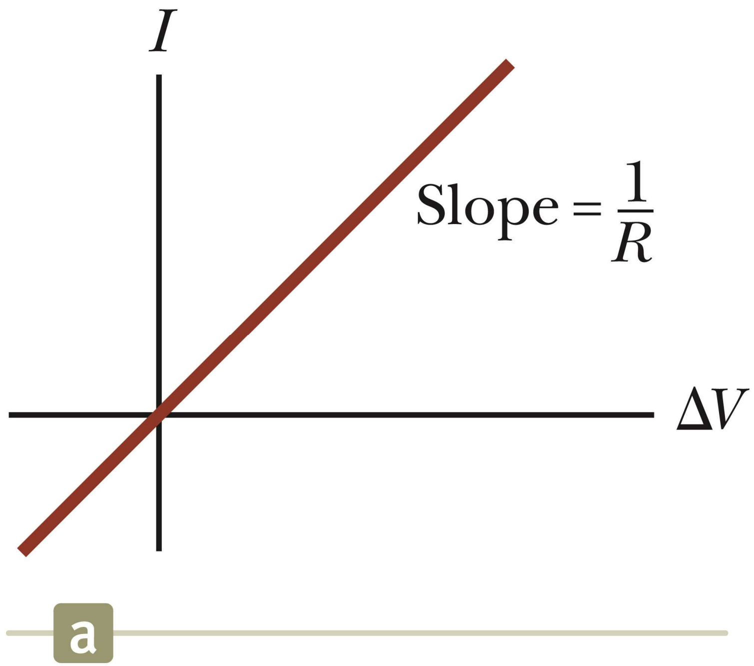


Resistance and Resistivity

- Every ohmic material has a characteristic resistivity that depends on the properties of the material and on temperature.
  - Resistivity is a property of substances.
- The resistance of a material depends on its geometry and its resistivity.
  - Resistance is a property of an object.
- An ideal conductor would have zero resistivity.
- An ideal insulator would have infinite resistivity.

Ohm's Law

- An ohmic device
- The resistance is constant over a wide range of voltages.
- The relationship between current and voltage is linear.
- The slope is related to the resistance.



Example 1:

- A steady current of 2.5 A exists in a wire for 4.0 min.

  (a) How much total charge passed by a given point in the circuit during those 4.0 min?

  (b) How many electrons would this be?

Solution

- a. Charge = current x time; convert minutes to seconds. Q = 600 C.
- b. Divide by electron charge: $n = 3.8 \times 10^{21}$ electrons.

Example 2:

- Calculate the resistance of an aluminum cylinder that is 10.0 cm long and has a cross-sectional area of 2.00×10^{-4} m^2. Repeat the calculation for a cylinder of the same dimensions and made of glass having a resistivity of 3.0×10^{10} Ω.m

Solution

$$R = \rho \frac{l}{A}$$

For aluminum, $\rho = 2.82x10^{-8}\,\Omega.\text{m}$

$$\therefore R = (2.82x10^{-8}\,\Omega.\text{m})\frac{(10.0x10^{-2}\,\text{m})}{(2.0x10^{-4}\,\text{m}^2)} = 1.41x10^{-5}\,\Omega$$

For Glass, $\rho = 3.0x10^{10}\,\Omega.\text{m}$

$$\therefore R = (3.0x10^{10}\,\Omega.\text{m})\frac{(10.0x10^{-2}\,\text{m})}{(2.0x10^{-4}\,\text{m}^2)} = 1.5x10^{13}\,\Omega$$

Example 3:

- Calculate the resistance per unit length of a 22-gauge Nichrome wire, which has a radius of 0.321 mm.
- If a potential difference of 10V is maintained across a 1.0-m length of the Nichrome wire, what is the current in the wire?

Solution

The cross sectional area of the wire is $A = \pi r^2 = \pi(0.321x10^{-3})^2 = 0.324x10^{-6}\ \text{m}^2$

$\rho = 1.5x10^{-6}\,\Omega.\text{m}$

$$\therefore \frac{R}{l} = 1.5x10^{-6}\,\Omega.\text{m}\frac{1}{(0.324x10^{-6}\ \text{m}^2)} = 4.6\Omega/\text{m}$$

$$I = \frac{\Delta V}{R} = \frac{10.0}{4.6x1.0} = 2.2\ \text{A}$$